# A Reimplementation & upgradation of ProsePose with SAM 3D Body

Konatala Vibhuvan Krishna(22024012), Atharva Sachin Gupta(22024006)

[1]*School of Bio-Medical Engineering, Indian Institute of Technology (BHU), Varanasi, Uttar Pradesh, India*

**CSO-432 AI for programmers**

# Objective & Problem Definition

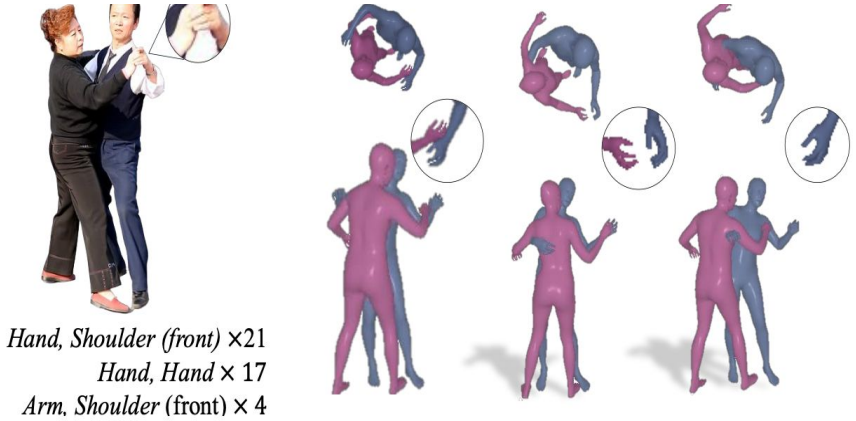
**Goal:** Estimate accurate 3D human pose from a single image

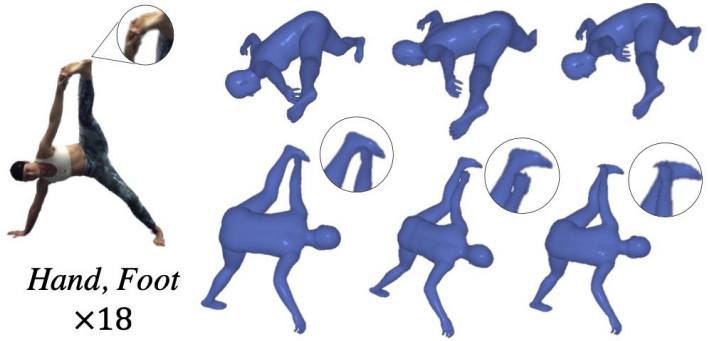
**Core Challenges:**

1. Occlusion: body parts hidden during interaction
2. Self-contact: complex poses like yoga, hugging
3. Depth ambiguity: multiple valid 3D interpretations
4. Standard HMR models process each person independently

**Resulting Issues:**

1. Meshes float apart where bodies visibly touch
2. Meshes penetrate each other (physically impossible)
3. Missing or incorrect contact regions



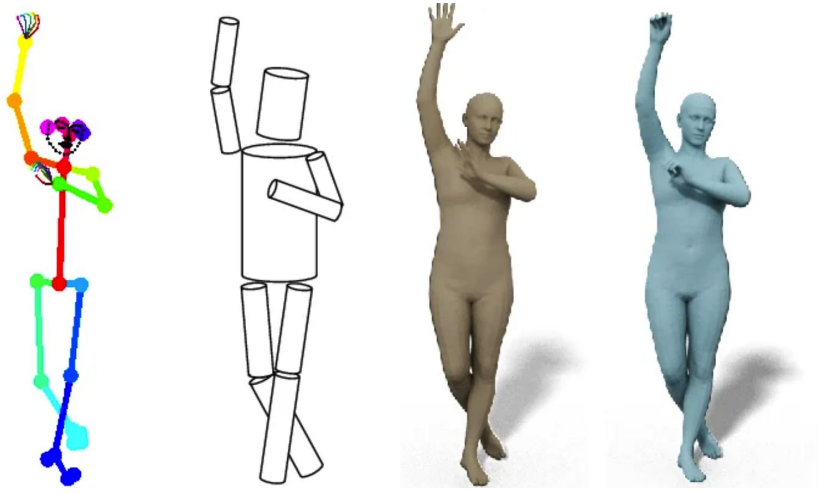




**Takeaway:** Geometry alone is insufficient to capture human contact and interaction

# Motivation & Gaps

**MOTIVATION:** 3D human interaction understanding is needed for:

1. Sports and performance tracking
2. Healthcare: posture and gait analysis
3. AR/VR and animation
4. Human-robot interaction



**EXISTING APPROACHES & THEIR GAPS:**

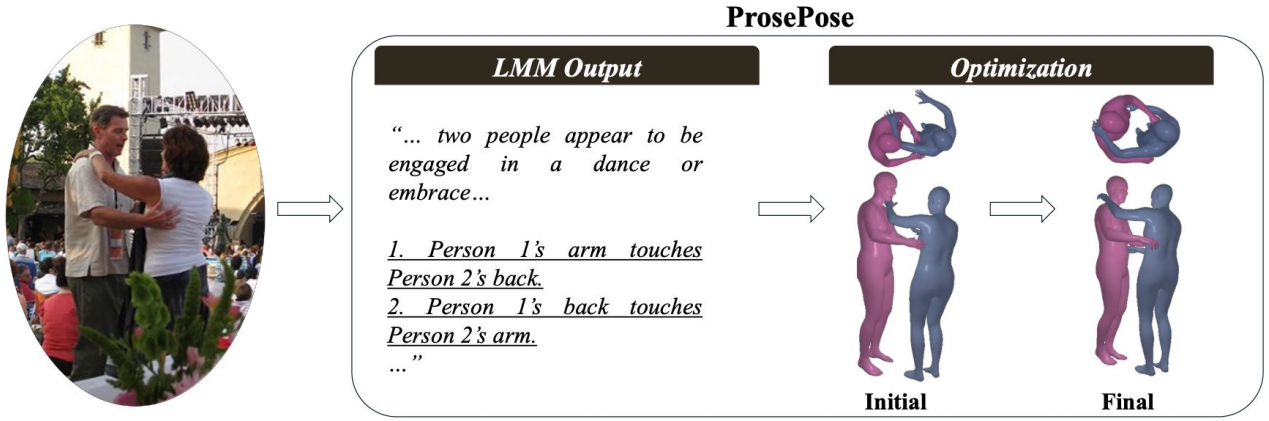
- ❖ Supervised contact-aware methods (e.g. BUDDI):
  Require expensive, scarce 3D contact annotations

- ❖ Optimization-based refinement (pose priors, heuristics, interpenetration penalties):
  Rule-based, do not capture semantic interaction

- ❖ Multi-person pose regressors (BEV, HMR2):
  Give good initial mesh but do not solve contact and also Heavy C++ dependencies, hard to reproduce

**KEY GAP:** No method exploits VLMs to derive contact constraints from images without any 3D annotation

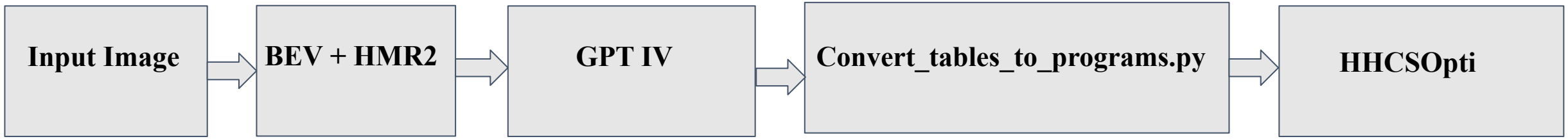
# ProsePose: The original Paper




**CORE IDEA:**

Use a Vision-Language Model (VLM) as a contact prior

Language → Mathematical Constraints → Optimized Pose




**KEY INSIGHT:**

Even imperfect VLM outputs improve final pose quality,  No 3D contact annotations needed at any stage

# Why We Replaced BEV + HMR2

PROBLEM WITH ORIGINAL INITIALISER:
i. BEV requires a compiled C++ renderer
ii. HMR2 needs a specific Detectron2 build
iii. Not reproducible on Windows / modern CUDA
iv. Dependency conflicts - setup takes days

WHY SAM 3D BODY:
i. State-of-the-art single-image mesh reconstruction
ii. Installs cleanly via pip — no C++ build at all
iii. Promptable: takes YOLOv8 bounding boxes as input
iv. Actively maintained, HuggingFace hosted

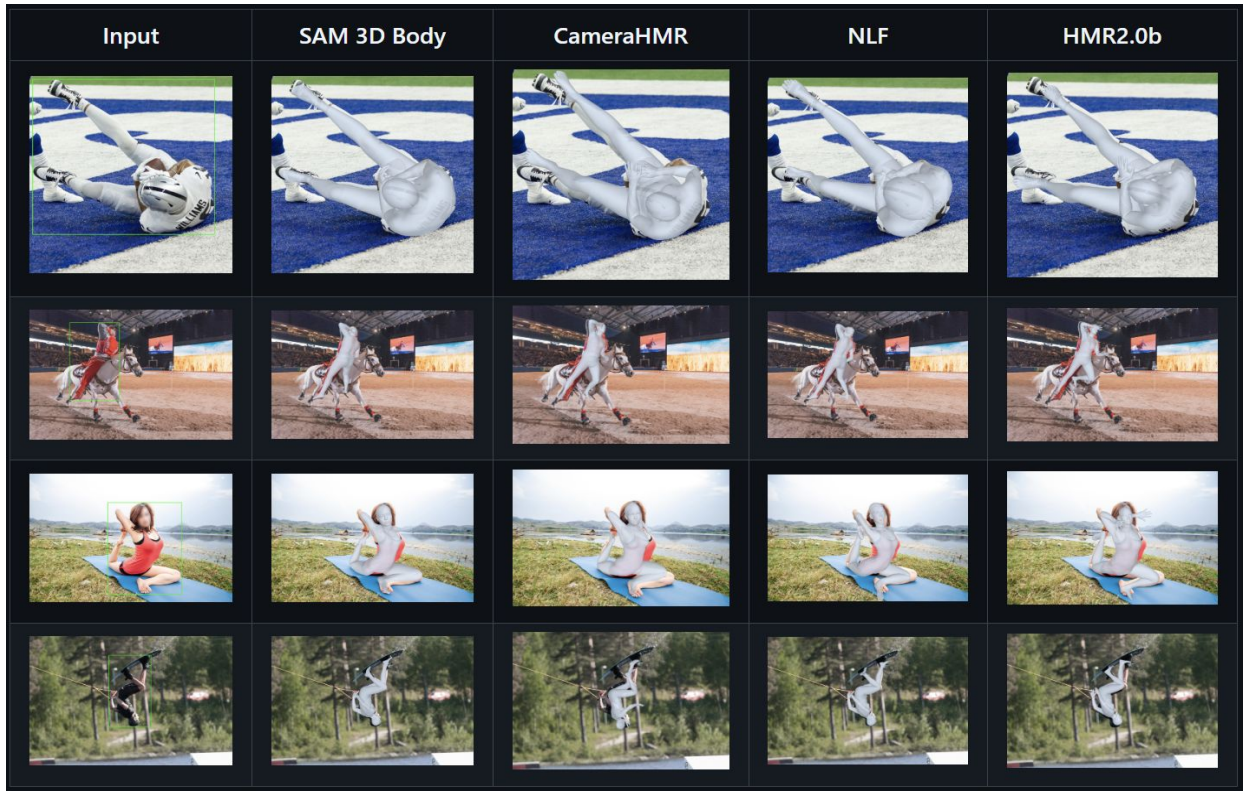


Image: Qualitative Results

WHAT WE KEPT UNCHANGED:
i. convert_tables_to_programs.py
ii. HHCSOpti optimization module
iii. Full ProsePose llib library

WHAT WE CHANGED:
i. BEV + HMR2 → SAM 3D Body
ii. GPT-4V → Llama 3.2 Vision (OpenRouter)
+ Gemini 2.0 Flash (fallback 1) + GPT-4o (fallback 2)

# Our Pipeline

STAGE 1 - MESH INITIALISATION:

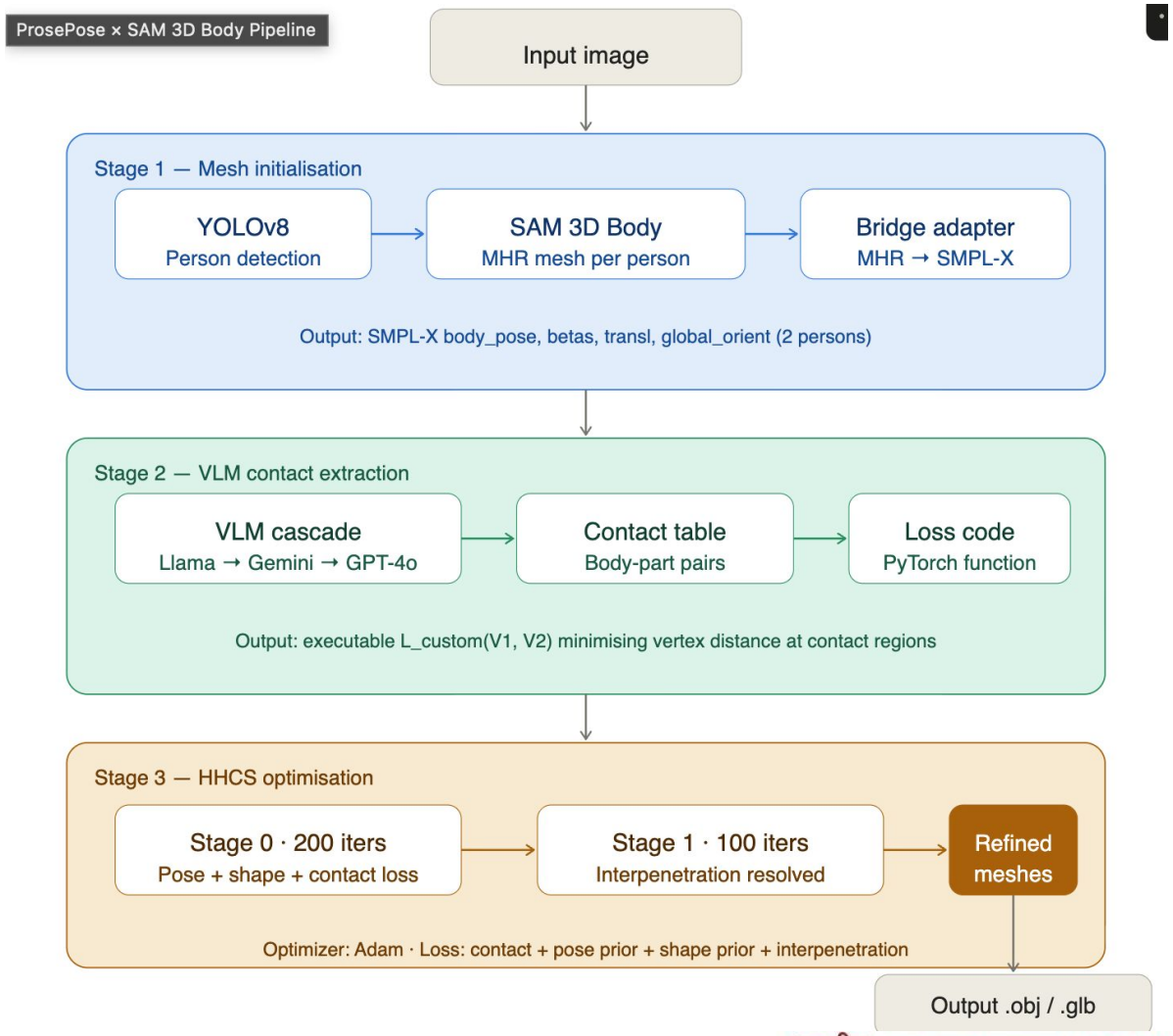
i. YOLOv8 detects both persons → bounding boxes
ii. SAM 3D Body reconstructs per-person 3D mesh
iii. Output: MHR-format body parameters

STAGE 2 - VLM CONTACT EXTRACTION:

i. Image sent to VLM with fixed prompt
ii. VLM returns contact table (body-part pairs)
iii. Table → executable PyTorch loss function

STAGE 3 - HHCS OPTIMISATION:

i. downsampled the meshes to improve speed of loss function calculation
ii. Stage 0: coarse contact alignment (200 iters)
iii. Stage 1: interpenetration resolution (100 iters)
iv. Output: refined SMPL-X meshes for both persons



# The MHR → SMPL-X Bridge

**THE PROBLEM:** SAM 3D Body outputs MHR format HHCSOpti expects SMPL-X format A conversion layer is needed

**KEY DIFFERENCES:**

| MHR | SMPL-X |
|---|---|
| 23 body joints → | 21 body joints |
| Rotation matrices → | Axis-angle vectors 63-dimensional vector |
| (N, 3, 3) shape | |
| MHR shape params → β (10-dim) | |

WHAT OUR BRIDGE DOES (integrated_bridge.py):

- Detects rotation matrix format
- Converts via matrix_to_axis_angle (PyTorch3D)
- Selects first 21 joints → flattens to 63-dim
- Zero-pads shape to 10 dimensions
- Returns clean init_human dict for HHCSOpti

# Loss Function & Optimisation

**CONVERTING LANGUAGE TO LOSS:**

1. VLM identifies: "hand touches shoulder"
2. Code maps body-part name → SMPL-X vertex region
3. Loss = minimum vertex distance between those regions
4. If contact pair identified → minimize that distance

TOTAL LOSS:

L = contact loss (VLM-derived, always active)
+ init pose loss (stay close to SAM3D init)
+ GMM pose prior (penalise unnatural poses)
+ shape prior (regularise body shape)
+ interpenetration (resolve mesh overlaps)

TWO-STAGE STRATEGY:

Stage 0: Establish coarse contact shape + pose optimised together
Stage 1: Resolve penetrations shape frozen, interpenetration loss ON

# Limitations:

1. Role of the LLM in the Pipeline The contact constraint generation step relies entirely on a Vision-Language Model (LLM) to interpret physical interactions between people, such as handshakes, embraces, or collisions, directly from the input image. These interpreted constraints are then converted into differentiable PyTorch loss functions that guide the 3D mesh optimisation. The quality of the final reconstructed pose is therefore directly proportional to the semantic accuracy of the LLM's description.

2. Free-Tier Model Underperforms on Spatial Reasoning The currently integrated model (via OpenRouter) frequently produces vague, generic, or geometrically incorrect contact descriptions. It struggles with precise spatial language (e.g., *"left hand of Person A is touching the right shoulder of Person B"*), leading to poorly targeted loss functions that either do not converge or pull meshes in incorrect directions during optimisation.

# Limitations:

3Advanced vision-language models like GPT-4o (Vision), Claude 3.5 Sonnet, and Gemini 1.5 Pro are paywalled, making constraint quality a major source of error. Incorrect constraints misguide the contact loss, causing the Adam optimiser to minimise the wrong objective and leading meshes away from true interactions, with repulsion and regularisation unable to fully correct it.

4.The loss function is supposed to work on SMPL-X but we are getting the mesh in MHR due the fundamental difference in the formats of these two models. The loss function is performing substandardly

5. Potential Solutions & Future Work

- Upgrade LLM access: Integrate GPT-4o or Gemini 1.5 Pro with proper API billing for higher-fidelity constraint generation.
- Few-shot prompting: Provide the LLM with curated examples of correct contact JSON to reduce hallucination and improve consistency.
- Fine-tuned specialist model: Train a lightweight vision model specifically on body-contact classification to replace the general-purpose LLM entirely.
- Constraint validation layer: Add a geometric sanity-check that filters out LLM-generated constraints that are physically impossible given the detected bounding boxes.

# Results & Observations

WHAT THE IMAGES SHOW:

1. Step 0: Total Loss = 14.18 — meshes heavily overlapping
2. Step 25: Total Loss = 0.057 — separation begins
3. Step 50: Total Loss = 0.057 — poses stabilising
4. Step 75: Total Loss = 0.053 — meshes cleanly separated
5. Step 100: Total Loss = 0.050 — individual poses clear
6. Step 125: Total Loss = 0.047 — final refined state

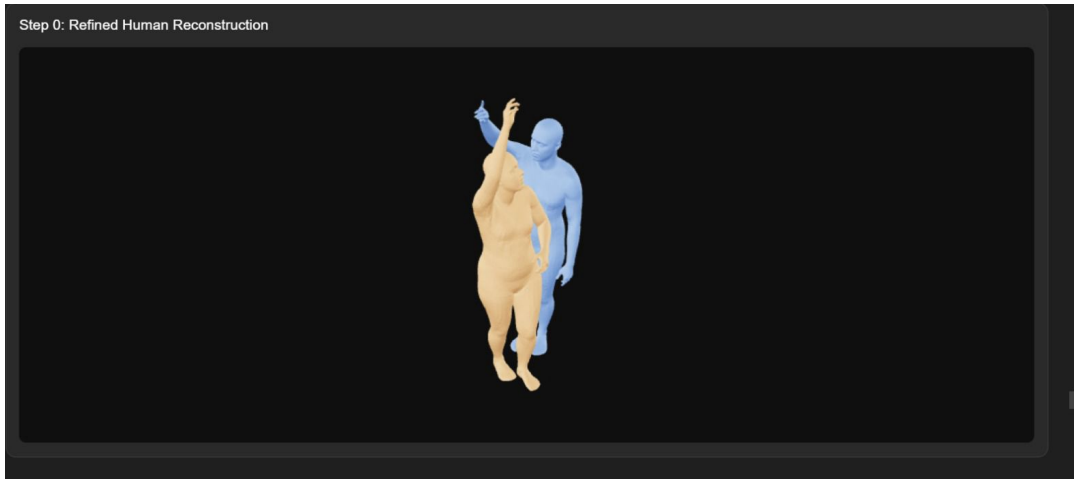
KEY OBSERVATION:

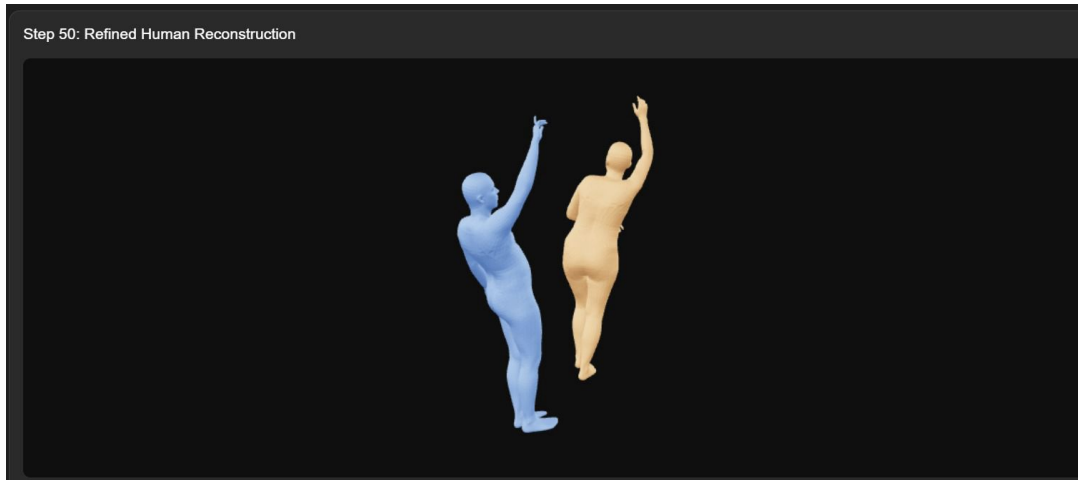
- Loss dropped from 14.18 → 0.047 over 125 steps
- Repel loss reaches 0.000 — no more interpenetration
- Meshes go from fully overlapping to physically plausible separated poses
- Pose quality (arms raised, body shape) preserved throughout optimisation

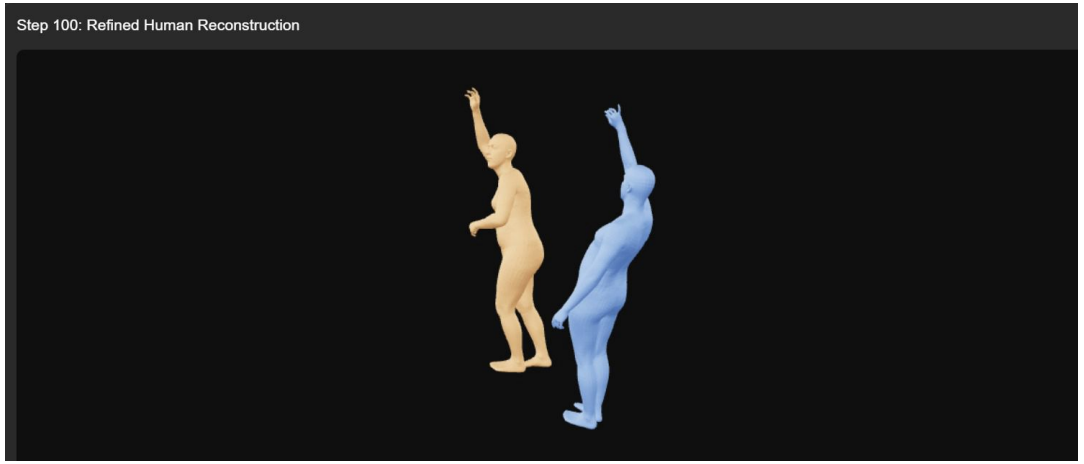
VLM FALLBACK CHAIN:

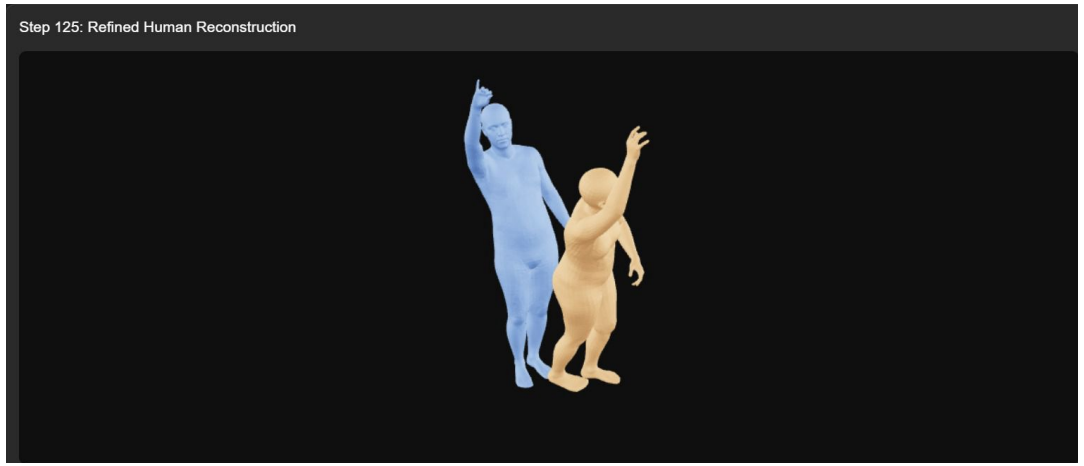
1: Llama 3.2 Vision (OpenRouter) 2: Gemini 2.0 Flash (fallback 1) 3: GPT-4o (fallback 2)

# Results & Observations



# Conclusion & Future Scope

**CONCLUSION:**

- Successfully reimplemented ProsePose with SAM 3D Body as a clean drop-in for BEV + HMR2
- Eliminated all C++ build dependencies
- Built MHR → SMPL-X adapter (integrated_bridge.py)
- Three-level VLM fallback chain ensures robustness
- Full pipeline in a single Jupyter notebook on a consumer GPU, no special setup needed

**LIMITATIONS:**

- MHR → SMPL-X shape mapping is approximate
- 2D keypoint loss disabled (no OpenPose from SAM3D)
- Fixed camera: no FOV estimation used
- VLM can hallucinate contacts / confuse left/right

**FUTURE SCOPE:**

- Use SAM3D's MoGe2 FOV estimate to re-enable the 2D keypoint loss
- Train a proper MHR → SMPL-X shape regressor
- Weight VLM contact pairs by confidence score
- Quantitative eval on CHI3D / FlickrCI3D benchmarks
- Extend to more than two interacting people
- Real-time deployment for AR/VR applications